

Sales Data Analysis: Data Storytelling and Statistical validation

Task 4 - Data Analytics Internship

Presented by : Devangam Narayanaswamy Veena

College : Sri Venkateshwara College of Engineering, Tirupati

Branch : Information Technology

Date : July 2026

Project Objective

- Perform data cleaning and exploratory data analysis (EDA) to improve data quality.
- Use SQL to answer business questions through filtering, aggregation, and data analysis.
- Develop interactive dashboards in Power BI to visualize key performance indicators (KPIs).
- Present findings through data storytelling and statistical validation to support business decision-making.
- Analyze the sales dataset to identify trends, patterns, and business insights.

Dataset Overview

Dataset Name

- Sales Dataset

Description

- Retail sales transaction data
- Customer, product & sales details
- Used for sales performance analysis

Dataset Summary

- Records: 1000
- Columns: 11

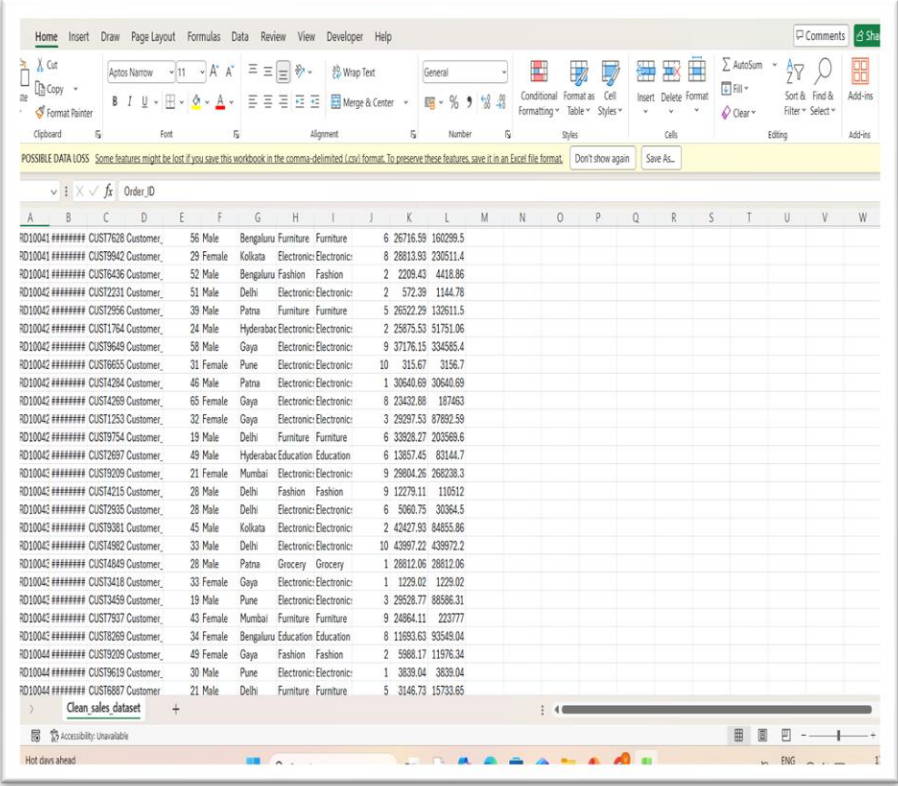
Key Columns

- Order_ID
- Customer_ID
- Product
- Category
- City
- Total_Sales

Order_ID	Customer_ID	Product	Category	City	Quantity	Unit Price	Total Sales
JRD1000C	CUST5529	Rice	Grocery	Bengaluru	7	2829.77	19808.39
JRD1000C	CUST3127	Book	Education	Bengaluru	5	27906.16	139530.8
JRD1000C	CUST8887	Book	Education	Bengaluru	8	37491.06	299928.5
JRD1000C	CUST2515	Mobile	Electronics	Kolkata	9	28541.36	256872.2
JRD1000C	CUST4796	Rice	Grocery	Bengaluru	10	14036.59	140365.9
JRD1000C	CUST3952	Laptop	Electronics	Hyderabad	9	35847.47	323527.2
JRD1000C	CUST1758	Mobile	Electronics	Patna	6	49997.53	299985.2
JRD1000C	CUST9622	Laptop	Electronics	Kolkata	3	9488.83	28466.49
JRD10001	CUST9770	Book	Education	Mumbai	6	19488.23	116929.4
JRD10001	CUST1312	Mobile	Electronics	Kolkata	8	18024.34	144194.7
JRD10001	CUST9642	Shoes	Fashion	Delhi	3	14151.42	42454.26
JRD10001	CUST3228	Chair	Furniture	Patna	6	47175.43	283052.6
JRD10001	CUST1038	Rice	Grocery	Patna	5	7413.68	37068.4
JRD10001	CUST6413	Shoes	Fashion	Patna	8	12042.52	96340.16
JRD10001	CUST8680	Mobile	Electronics	Patna	8	25465.31	203722.5
JRD10001	CUST9307	Shoes	Fashion	Delhi	10	20111.09	201110.9
JRD10001	CUST4363	Book	Education	Delhi	10	40109.12	401091.2
JRD10001	CUST3508	Shoes	Fashion	Gaya	1	7280.64	7280.64
JRD10002	CUST5806	Chair	Furniture	Mumbai	5	6122.59	30612.95
JRD10002	CUST6504	Laptop	Electronics	Kolkata	8	23770.36	190162.9
JRD10002	CUST8908	Shoes	Fashion	Pune	1	2344.98	2344.98
JRD10002	CUST2903	Book	Education	Mumbai	7	15396.37	107774.6
JRD10002	CUST1406	Shoes	Fashion	Gaya	10	31576.5	315765
JRD10002	CUST8036	Laptop	Electronics	Patna	2	11127.06	22254.12
JRD10002	CUST6532	Rice	Grocery	Pune	7	36069.86	252489

Data Cleaning Process

- Imported the sales dataset into MySQL.
- Checked for missing values and handled them.
- Removed duplicate records.
- Verified and corrected data types.
- Cleaned inconsistent values and standardized the dataset.
- Validated the cleaned data for accurate analysis.



POSSIBLE DATA LOSS: Some features might be lost if you save this workbook in the comma-delimited (.csv) format. To preserve these features, save it in an Excel file format. [Don't show again] [Save As...]

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
RD10041	#####	CUST7628	Customer,	56	Male	Bengaluru	Furniture	Furniture	6	26716.59	160299.5												
RD10041	#####	CUST9942	Customer,	29	Female	Kolkata	Electronic	Electronic	8	28813.93	230511.4												
RD10041	#####	CUST6436	Customer,	52	Male	Bengaluru	Fashion	Fashion	2	2208.43	4418.86												
RD10042	#####	CUST2231	Customer,	51	Male	Delhi	Electronic	Electronic	2	572.39	1144.78												
RD10042	#####	CUST2956	Customer,	39	Male	Patna	Furniture	Furniture	5	26522.29	132611.5												
RD10042	#####	CUST1764	Customer,	24	Male	Hyderabad	Electronic	Electronic	2	25675.53	51751.06												
RD10042	#####	CUST9649	Customer,	58	Male	Gaya	Electronic	Electronic	9	37176.15	334585.4												
RD10042	#####	CUST6655	Customer,	31	Female	Pune	Electronic	Electronic	10	315.67	3156.7												
RD10042	#####	CUST4284	Customer,	46	Male	Patna	Electronic	Electronic	1	30640.69	30640.69												
RD10042	#####	CUST4269	Customer,	65	Female	Gaya	Electronic	Electronic	8	23432.88	187453												
RD10042	#####	CUST1253	Customer,	32	Female	Gaya	Electronic	Electronic	3	29297.53	87892.59												
RD10042	#####	CUST9754	Customer,	19	Male	Delhi	Furniture	Furniture	6	33928.27	203569.6												
RD10042	#####	CUST2697	Customer,	49	Male	Hyderabad	Education	Education	6	13857.45	83144.7												
RD10042	#####	CUST9209	Customer,	21	Female	Mumbai	Electronic	Electronic	9	29004.26	268238.3												
RD10042	#####	CUST4215	Customer,	28	Male	Delhi	Fashion	Fashion	9	12279.11	110512												
RD10042	#####	CUST2935	Customer,	28	Male	Delhi	Electronic	Electronic	6	5060.75	30364.5												
RD10042	#####	CUST9381	Customer,	45	Male	Kolkata	Electronic	Electronic	2	42427.93	84855.86												
RD10042	#####	CUST4982	Customer,	33	Male	Delhi	Electronic	Electronic	10	43997.22	439972.2												
RD10042	#####	CUST4849	Customer,	28	Male	Patna	Grocery	Grocery	1	28812.06	28812.06												
RD10042	#####	CUST3418	Customer,	33	Female	Gaya	Electronic	Electronic	1	1229.02	1229.02												
RD10042	#####	CUST3459	Customer,	19	Male	Pune	Electronic	Electronic	3	29528.77	88586.31												
RD10042	#####	CUST7957	Customer,	43	Female	Mumbai	Furniture	Furniture	9	24964.11	223777												
RD10042	#####	CUST8259	Customer,	34	Female	Bengaluru	Education	Education	8	11953.63	95549.04												
RD10042	#####	CUST9209	Customer,	49	Female	Gaya	Fashion	Fashion	2	5988.17	11976.34												
RD10042	#####	CUST9619	Customer,	30	Male	Pune	Electronic	Electronic	1	3839.04	3839.04												
RD10042	#####	CUST6887	Customer,	21	Male	Delhi	Furniture	Furniture	5	3146.73	15733.65												

Clean_sales_dataset

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Hot days ahead

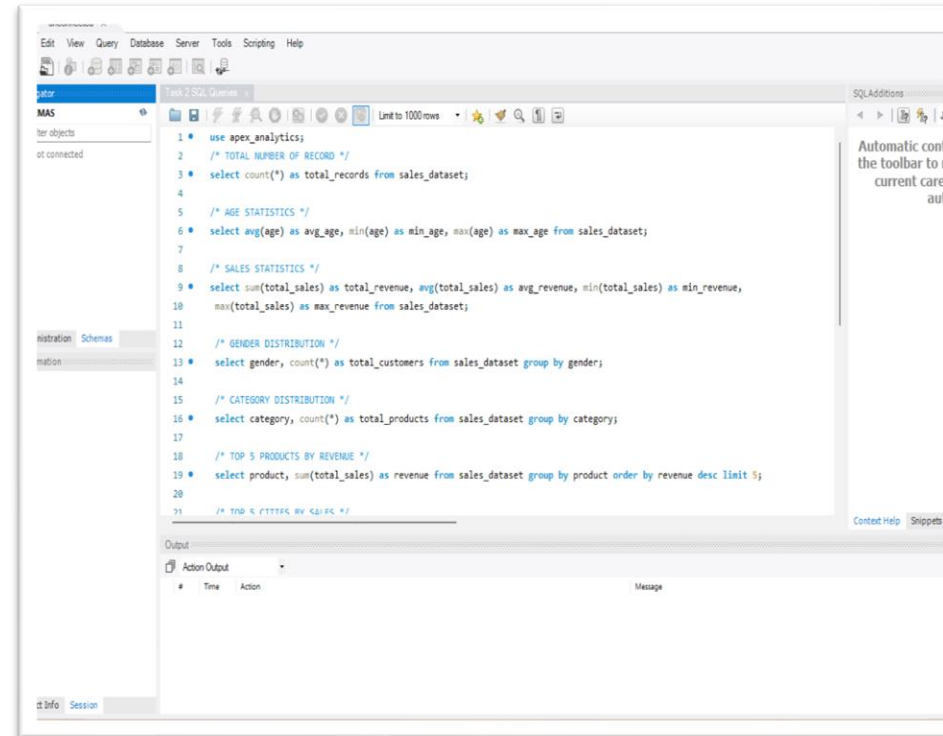
Exploratory Data Analysis(EDA) and SQL Analysis

Business Questions Answered

- Top 5 cities by total revenue.
- Best-selling product categories.
- Monthly sales trend analysis.
- Total revenue generated by each product.
- Customer distribution by gender.
- Category-wise sales performan.

Key Findings

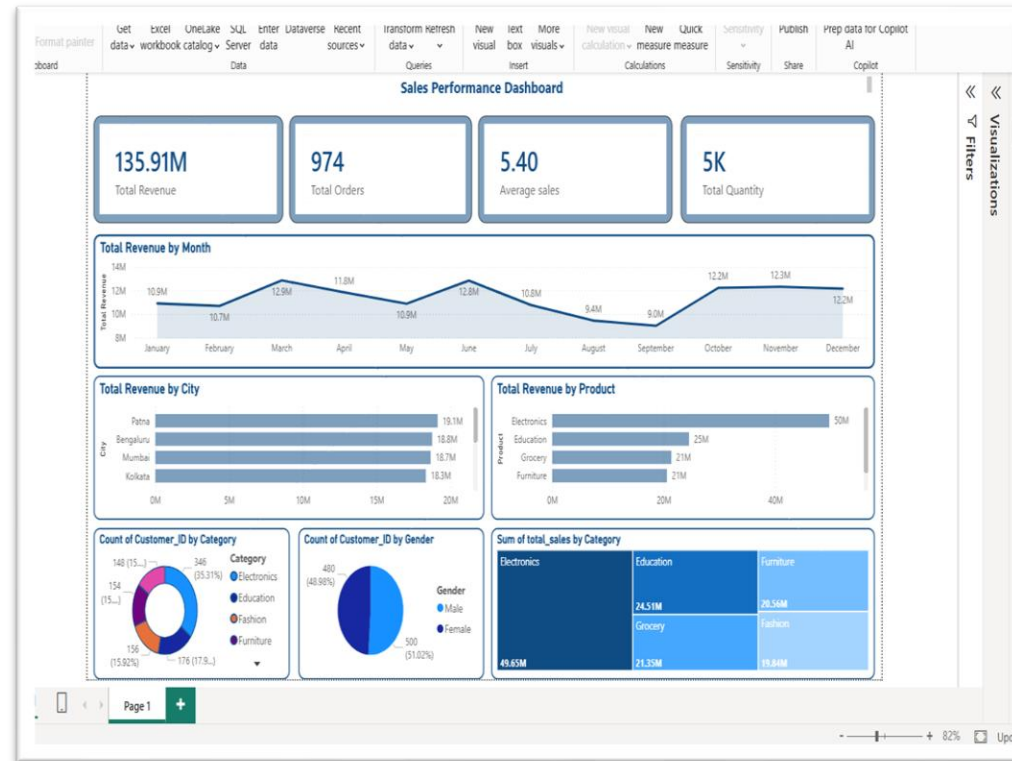
- Revenue varied significantly across cities.
- Electronics generated the highest sales.
- Monthly sales showed an overall increasing trend.
- Customer purchases were almost equally distributed by gender.



Power BI Dashboard

Dashboard Highlights

- Interactive dashboard developed using Power BI.
- KPI cards display Total Revenue, Total Orders, Average Sales, and Total Quantity.
- Visualizations show monthly sales trends, city-wise revenue, and product performance.
- Category-wise and gender-wise analysis helps understand customer purchasing patterns.
- Dashboard enables quick and data-driven business decision-making.



Key Business Insights

Key Insights

- Total revenue indicates strong overall sales performance across the dataset.
- A few cities contributed the highest share of total revenue.
- Electronics was the top-performing product category.
- Monthly sales analysis revealed clear trends in customer demand.
- KPI analysis helped monitor revenue, orders, average sales, and quantity sold.
- The dashboard provided valuable insights for business decision-making.

Business Recommendations

- Focus marketing efforts on top-performing cities to maximize revenue.
- Increase inventory for high-demand products and categories.
- Develop promotional strategies for low-performing products.
- Monitor monthly sales trends to improve sales planning and forecasting.
- Use the Power BI dashboard regularly to support data-driven business decisions.



Hypothesis Testing and Statistical Validation

Business Hypothesis

H_0 (Null Hypothesis): There is no significant difference in average sales between product categories.

H_1 (Alternative Hypothesis): There is a significant difference in average sales between product categories.

Statistical Test Used:

- Independent Sample **T-Test**

Conclusion:

- **The statistical test was used to evaluate the hypothesis and determine whether the observed difference in average sales was statistically significant.**
- **The results help support data-driven business decisions.**

Conclusion

Project Summary

- Successfully completed data cleaning, exploratory data analysis, SQL analysis, and Power BI dashboard development.
- Identified key business insights from the sales dataset through data analysis and visualization.
- Applied data storytelling and basic statistical validation to support business decision-making.
- Enhanced practical skills in SQL, Power BI, data analysis, and business intelligence.
- This project demonstrates the value of data-driven decision-making in solving real-world business problems.

Thank You!